Programming Project 1

Graded

Student

Jinku Jung

Total Points

60 / 60 pts

Autograder Score

0.0 / 0.0

Question 2

Problem 1 (Hours)

10 / 10 pts

Running

- ✔ **+ 6 pts** Program compiles and runs without error

Testing

- ✔ **+ 1 pt** Test 201, 0
- ✔ **+ 1 pt** Test 0, 25
- ✔ **+ 1 pt** Test 11, 13
- ✔ **+ 1 pt** Test 37, 1

please don't submit .class files

In future assignments please do not ask users for their name (unless the assignment specifically asks for it) as it interferes with grading.

Amazing!

Question 3

Problem 2 (Prime)

20 / 20 pts

Running

- ✔ **+ 12 pts** Program compiles and runs without error

Testing

- ✔ **+ 2 pts** Test 613
- ✔ **+ 2 pts** Test 51
- ✔ **+ 2 pts** Test 50061
- ✔ **+ 2 pts** Test 2

please don't submit .class files

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Amazing!

Question 4

Problem 3 (Easter)

30 / 30 pts

Running

- ✔ **+ 15 pts** Program compiles and runs without error

Test 1: 2023

- ✔ **+ 3 pts** Month or day correct
- ✔ **+ 2 pts** Both correct

Test 2: 1991

- ✔ **+ 3 pts** Month or day correct
- ✔ **+ 2 pts** Both correct

Test 3: 2099

- ✔ **+ 3 pts** Month or day correct
- ✔ **+ 2 pts** Both correct

please don't submit .class files

In future assignments please do not ask users for their name (unless the assignment specifically asks for it) as it interferes with grading.

Amazing!

Autograder Results

This assignment does not have an autograder configured.

Submitted Files

.codio

Download

```
{
  // This file is used to configure the three buttons along Codio's top menu.

  // Run button configuration
  "commands": {
    "Compile & Run": "javac {{filename}} && java {{filename_no_ext}}",
    "Compile": "javac {{filename}}",
    "Run": "java {{filename_no_ext}}"
  },
  // Preview button configuration
  "preview": {
    "Project Index (static)": "https://{{domain}}/{{index}}",
    "Current File (static)": "https://{{domain}}/{{filepath}}",
    "Box URL": "http://{{domain3000}}/",
    "Box URL SSL": "https://{{domain3000}}/"
  },
  // Debugger target button configuration
  "debugger": [{
    "type": "JAVA",
    "command": "-agentlib:jdwp=transport=dt_socket,server=y,suspend=y,address=5105",
    "before": "javac -g -d . {{filepath}} && java -
agentlib:jdwp=transport=dt_socket,server=y,suspend=y,address=5105 {{filepath}} ",
    "single": true,
    "path": "{{filepath}}",
    "args": "",
    "rootFolder": "",
    "uuid": "17619cd9-b8af-656f-9f29-54ca4d9f5d4a",
    "name": "Debug Current File"
  }]
}
```

Easter.class

Download

```
Binary file hidden. You can download it using the button above.
```

Easter.java   Download

```java
/*
 * This program implements the algorithm created by [Carl] Fredrich Gauss
 * for determining the date of Easter
 *
 * @author: Jinku Jung
 * @date: 1/30/2021
 */

import java.util.Scanner;

public class Easter{

    public static void main(String[] args){

        //Initializing Scanner class variable "input"
        Scanner input = new Scanner(System.in);
        System.out.println("Hello!  Please enter your desired year for the Easter holiday.");

        //Declaring and initializing all necessary dummy variables and literals
        int y = 0; //Independent variable
        int a, b, c, d, e, g, h, j, k, m, n, r, p = 0;  //Dependent variables
        String thanks = "Thank you for using my Easter Calendar program!  --Jinku Jung";

        //Calculations to determine the month and day of Easter of the user's *year*

            //Gauss' Algorithm (Step 1)
            y = (int) input.nextInt();

            //Gauss' Algorithm (Step 2)
            a = (y % 19);

            //Gauss' Algorithm (Step 3)
            b = (y / 100);
            c = (y % 100);

            //Gauss' Algorithm (Step 4)
            d = (b / 4);
            e = (b % 4);

            //Gauss' Algorithm (Step 5)
            g = (8 * b + 13) / 25;

            //Gauss' Algorithm (Step 6)
            h = (19 * a + b - d - g + 15) % 30;

            //Gauss' Algorithm (Step 7)
            j = (c / 4);
            k = (c % 4);

```

```
        //Gauss' Algorithm (Step 8)
        m = (a + 11 * h) / 319;

        //Gauss' Algorithm (Step 9)
        r = (2 * e + 2 * j - k -h + m +32) % 7;

        //Gauss' Algorithm (Step 10)
        n = (h - m + r + 90) / 25;

        //Gauss' Algorithm (Step 11)
        p = (h - m + r + n + 19) % 32;

      //Printing out the final result
      System.out.println("Easter will occur on " + n + "/" + p + " in the Gregorian Calendar, during the year " + y + " AD." + "\n" + thanks);
  }
}

/*
* Coder's Note: This program was made possible through the sonorous classic rock of WNCX via Mini Radio Player for Windows 10, VS  * Code (kudos for its high contrast a * la Brackets.io for HTML), and one 8.4 oz can of Red Bull Sugar-Free.  It is WAY more fun to code on an a w e s o m e Sony Bravia X950G 55" 4K TV while sitting       * recumbent in a leather recliner than a "decent" Dell Ultrasharp 34" UltraWide hunched over in a crummy AKRacing Gaming Chair from Best Buy.  Lesson Learned?  Don't * believe the WFH hype!  --Jinku
*/
```

Hours.class Download

```
Binary file hidden. You can download it using the button above.
```

Hours.java Download

```java
/**
 * This program converts a number of days and weeks into the
 * equivalent number of hours
 *
 * @author: Jinku Jung
 * @date: 1/30/2021
 */

import java.util.Scanner;

public class Hours{

   public static void main(String[] args){

      //Calls to Scanner.  Adapted from "Example4.java" (Cannon)
      Scanner input = new Scanner(System.in);
      System.out.println("Please enter your name.");
      String name = input.nextLine();

      //Declare all variables and constants
      int days;
      int weeks;
      final int DAYS_IN_A_MILLENIUM = 365000;
      final int WEEKS_IN_A_MILLENIUM = 52000;
      String thanks = "Thank you for using my time conversion program!  --Jinku Jung";

      //Grab the inputs from the user
      System.out.println("Please enter a whole number amount of days, " + name + ".");
      days = input.nextInt();

      System.out.println("Please enter a whole number amount of weeks, " + name + ".");
      weeks = input.nextInt();

      //Establish *reasonable* upper bounds for the user's input values
      while(days > DAYS_IN_A_MILLENIUM || weeks > WEEKS_IN_A_MILLENIUM)
      {
         System.out.println("Please enter a reasonable value for days.");
         days = input.nextInt();
         System.out.println("Please enter a reasonable value for weeks.");
         weeks = input.nextInt();
      }

      //Program calculations
      int hours = (days * 24);
      hours = hours + (weeks * (24 * 7));

      //Program output or results
      System.out.println(days + " days and " + weeks + " weeks yield " + hours + " total hours." + "\n" + thanks);
```

```
    }
}
```

Prime.class Download

Binary file hidden. You can download it using the button above.

Prime.java Download

```java
/*
 * This program accepts a positive integer as input and
 * determines whether or not the integer is prime.
 *
 * @author: Jinku Jung
 * @date: 1/30/2021
 */

import java.util.Scanner;

public class Prime{

    public static void main(String[] args){
        {
            // Initializing Scanner class variable "input"
            Scanner input = new Scanner(System.in);

            //User salutaions
            System.out.println("Hello!  Please enter your name.");
            String name = input.nextLine();

            //Grab and store the user's candidate prime number
            System.out.println("Now enter your candidate prime number, " + name + ".");
            int candidatePrime = 0;
            candidatePrime = input.nextInt();

            //Program calculations
            int evenPrimeFactor = 2;
            int oddPrimeFactor = 3;
            boolean isNotPrime = false;
            boolean isPrime = false;
            String thanks = "Thank you for using my Prime Number java program!  --Jinku";

            //Test for the special prime number case of 2
            if (candidatePrime == evenPrimeFactor)
            {
             isPrime = true;
             System.out.println("Your number IS prime." + "\n" + thanks);

            }

            //Eliminate all even-numbered factors of candidatePrime
            if ((candidatePrime % evenPrimeFactor == 0) && (isPrime != true))
            {
              System.out.println("Your number is NOT prime." + "\n" + thanks);
              isNotPrime = true;
            }

            //Test for all other natural numbers that are odd-numbered factors of candidatePrime
```

```java
            while ((candidatePrime != 0) && ((oddPrimeFactor * oddPrimeFactor) <= candidatePrime) && ((isNotPrime != true) && (isPrime != true)))
            {
                if (candidatePrime % oddPrimeFactor == 0 )
                {
                        System.out.println("Your number is NOT prime." + "\n" + thanks);
                        isNotPrime = true;
                }
                oddPrimeFactor += 2;
            }
            if ((isNotPrime == false) && (isPrime != true))
            {
                isPrime = true;
                System.out.println("Your number IS prime." + "\n" + thanks);
            }
        }
    }
}

//Coder's Note: Cut & pasted my last code revision of Prime.java from VS Code to debug more effectively.  LOTS of time spent removing all "return" statements from this programming assignment, per Canon's rubric.  Thank god for truth tables and Boolean logic!  --Jinku
```

README.md Download

COMS W1004
Programming Project 1
Due February 1 at 11:59PM
____

PROGRAMMING (60 points)

1. (10 points) Create an application in Java that asks a user for a number of days and weeks and computes the equivalent number of hours. To do this, edit the file Hours.java included here.

2. (20 points) Implement your algorithm from problem 20 in Chapter 2 of Schneider and Gersting. That is, write a Java program that executes your algorithm. To do this edit the file Prime.java included here.

3. (30 points) Do programming project P4.2 in Big Java: Early Objects 7th Edition. Edit the file Easter.java included here.

Recommended: *(not for credit):* Programming exercises and projects P4.5, P4.10, E5.12, E5.16, P5.1, P5.10 in the 7th Edition of ***Big Java Early Objects***.

What to hand in
You should submit three files for this assignment: Hours.java, Prime.java, and Easter.java

Submitting your work:
Please submit your work electronically via Gradescope. To do this, export your .java files from Codio, compress them into a single zip file; please name this file [your UNI here]_pp1.zip (without the brackets). So I would submit a single file named ac1076_pp1.zip Submit this file under Programming Project 1. No need to tag anything for the .java files.